MINURI

Your starter guide to independent living

Iteration 1 — Analysis & Design Report

FIT5120 Industry Experience — 2026 Semester 1
Topic: Shielding Crisis & Community Resilience
SDG 3: Good Health and Wellbeing | SDG 11: Sustainable Cities and Communities

Scope: Melbourne, Victoria

Team [Your Team Name] — Team [Number]
Submitted: Week 5, 2026 S1

1. Introduction

This document defines the analysis and design specifications for Iteration 1 of Minuri — a proactive wellbeing web application for young Australians aged 18–25 living independently for the first time in Melbourne. It outlines the problem context, target audience, feature scope, user stories with acceptance criteria, open data strategy, and technical approach. This report informs the Iteration 1 build and will be referenced throughout development and review.

2. Project Overview

2.1 Problem Statement

Young adults aged 18–25 transitioning into independent living for the first time — including university students relocating from regional areas and young professionals starting their first role in Melbourne — experience a significant and underrecognised wellbeing gap upon leaving home. Without the informal support of family life — shared meals, ambient check-ins, daily routine nudges — many struggle with eating, health, social connection, and navigating a new city. A 2024 Orygen report found that young adults living away from home for the first time are significantly more likely to experience psychological distress, with many citing a lack of informal support as a primary barrier to seeking help. Existing digital tools fail this context: they are too clinical, too reactive, or too generic. Minuri provides the practical knowledge and locally relevant Melbourne resources that a parent or housemate would have delivered informally — before a crisis occurs.

2.2 Target Audience

Minuri's primary audience is Australian young adults aged 18–25 living independently for the first time in Melbourne. This spans two sub-groups: the university student who has relocated from a regional area and is navigating uni life, share houses, and a new city simultaneously; and the young professional who has moved cities for work and no longer has the structure of a student community. What unites them is a shared experience: the informal routines and oversight that family life provided have disappeared overnight.

News Reference

The Educator (March 2026). 'What are Australia's youth thinking and feeling in 2026?' — Mission Australia's 2025 Youth Survey of 17,000+ young people identified cost of living (64%), mental health (29%), and housing (25%) as top national concerns.

2.3 Personas

Two personas ground design decisions. Each maps directly to Minuri's features through a "problem → feature → outcome" table, ensuring nothing is built without a clear user need.

Persona 1: Alex — The Uni Student	
Name & Age	Alex, 19
Background	Domestic student from Bendigo, regional Victoria. First year at Monash Clayton. Sharing a house with two students he barely knows from O-Week. Mum handled all meals, appointments, bills, and routines at home. Now in Melbourne for the first time — surrounded by people but completely on his own.

Goals	Get through first year without burning out. Figure out how to feed himself on a budget. Make friends in a city where he knows no one. Stop calling home every time something feels hard.
Pain Points	Has Medicare but has never booked a GP. Doesn't know what bulk-billing means. Eats the same $14 pad thai daily. Doesn't understand Myki zones or the free tram zone. Sits alone on weekends because he can't afford to go out and doesn't know what's free. Googles "am I depressed or just lonely" at 11pm.
Quote	"I know I need help with stuff. I just don't know what to Google."

How Minuri helps Alex:

Problem	Minuri Feature	What They Get
Doesn't know what to buy or cook	**Guides → Eating & Cooking**	"Your First Grocery Run" checklist + "5 Meals With 5 Ingredients"
Eats the same meal every day	**Guides → "Cheap Eats" + data widget**	Interactive map of cafés near campus from CoM CLUE data
Doesn't understand Myki or zones	**Guides → Getting Around**	"Melbourne Transport 101" — Myki, concessions, free tram zone
Never booked a GP	**Guides → Health & Wellbeing**	"How to Book Your First GP" with a phone script
Doesn't know where the nearest GP is	**Near Me → Health filter**	Enters "Clayton" → bulk-billing GPs on map (NHSD + CoM data)
Bored and broke on weekends	**Guides → "Free Things to Do" + data**	Parks, BBQs, galleries from CoM datasets
Lonely, feels like the only one	**Guides → "Feeling Lonely?" + CoMSIS**	Real loneliness data from CoM Social Indicators
In crisis	**Crisis Lines (always visible)**	Lifeline, Beyond Blue, headspace — every page

Persona 2: Jordan — The First Jobber	
Name & Age	Jordan, 23
Background	Just started first full-time role as a junior developer at a tech company in Melbourne CBD. Moved from Brisbane three weeks ago. Renting a studio in Southbank alone. No longer has the structure of uni — no orientation, no campus map, no cohort. All friends are in another city.
Goals	Perform well at work without burning out. Build a routine and social life from scratch. Finally register with a GP instead of ignoring health problems.
Pain Points	Defaults to the same sushi train because it's the only place they recognise. Keeps putting off finding a GP. Orders Uber Eats instead of cooking. Wants to find a bar or meetup but doesn't know where locals go. Feels guilty for not exercising, cooking, or socialising — but too drained to act.
Quote	"I track every KPI at work. I don't track a single thing about my own life."

How Minuri helps Jordan:

Problem	Minuri Feature	What They Get
Eats the same meal every night	**Near Me → Food & Dining**	Every café/restaurant in Southbank from CoM CLUE data
Wants to find a bar nearby	**Near Me → Bars & Social**	Bars within walking distance with patron capacity
Keeps putting off finding a GP	**Guides → "First GP" + Near Me Health**	3-min guide → "Find a GP" → bulk-billing clinics in Southbank/CBD
Should talk to someone about stress	**Guides → "Bulk-Billing" + Near Me Mental Health**	Discovers MHCP gives 10 subsidised sessions. headspace + psychologists nearby
Never learned to cook	**Guides → Eating & Cooking**	"5 Meals With 5 Ingredients" + "First Grocery Run"
Wants to exercise outdoors	**Near Me → Parks & Free**	Parks, sports facilities, walking paths near Southbank
Needs amenities while exploring	**Near Me → Amenities**	Public toilets + drinking fountains with accessibility info
In crisis — burnout, isolation	**Crisis Lines + guide**	Beyond Blue, headspace, Lifeline — one tap away

3. Current Iteration

3.1 High-Level Solution

Iteration 1 delivers two core features addressing both sides of the pre-crisis gap: the knowledge gap (not knowing what you need to know) and the action gap (not knowing where to go). Both require no authentication, use 9 open datasets, and form a complete user journey. Scope: Melbourne only.

Feature 1: First Time Guides — 15 practical guides across 5 categories. Selected guides embed live data widgets from CoM APIs.

Category	Guide Title	Open Data Used	Type
🍳 **Eating**	Cheap Eats in Melbourne	CoM Cafés API	Data-enhanced
🍳 **Eating**	Your First Grocery Run	CoM Landmarks (Retail)	Data-linked
🍳 **Eating**	5 Meals With 5 Ingredients	—	Static
🚋 **Transport**	Melbourne Transport 101	—	Static
🚋 **Transport**	How to Get From A to B	—	Static
🏥 **Health**	Book Your First GP	NHSD + CoM Landmarks	Data-linked
🏥 **Health**	Bulk-Billing Explained	Near Me GP finder	Data-linked
🏥 **Health**	GP vs Emergency	— (crisis lines shown)	Static

🏠 **Adulting**	First Week Living Alone Checklist	Near Me (all filters)	Data-linked
🏠 **Adulting**	Share House Survival Guide	—	Static
💙 **Social**	Free Things to Do in Melbourne	CoM Landmarks + BBQs	Data-enhanced
💙 **Social**	Feeling Lonely? You're Not Alone	CoM Social Indicators	Data-enhanced
💙 **Social**	Crisis Lines & Free Support	— (hardcoded)	Static

Feature 2: Near Me — Map-based Melbourne resource finder with 7 filter categories, aggregating CoM + NHSD + ABS data.

Category	Data Source	Displayed
Health	CoM Landmarks (Health) + NHSD	GPs, hospitals, bulk-billing flag
Mental Health	NHSD + headspace locations	headspace, psychologists, crisis lines
Food & Dining	CoM Cafés & Restaurants (CLUE)	Name, cuisine, seats, address
Bars & Social	CoM Bars & Pubs (CLUE)	Name, patron capacity, address
Groceries	CoM Landmarks (Retail) + curated list	Coles, Woolworths, Aldi, IGA
Parks & Free	CoM Landmarks + BBQs + Fountains	Parks, sports, BBQs, fountains
Amenities	CoM Toilets + Fountains	Toilets (accessibility info), fountains

3.2 Open Data Strategy

9 datasets total. 7 from CoM Open Data Portal (Opendatasoft API v2.1, no key), 1 ABS Census, 1 NHSD.

CoM API: https://data.melbourne.vic.gov.au/api/explore/v2.1/catalog/datasets/{id}/records

Dataset	Source	Usage	Insight
Landmarks	CoM	Near Me: Health, Parks, Retail filters	Maps service clusters and coverage gaps
Cafés (CLUE)	CoM	Near Me: Food. "Cheap Eats" data widget	Food accessibility by area, 2002–2023 trends
Bars (CLUE)	CoM	Near Me: Bars & Social	Social venue density
Public Toilets	CoM	Near Me: Amenities filter	Accessibility coverage
Drinking Fountains	CoM	Near Me: Amenities filter	Free hydration access
Public BBQs	CoM	Near Me: Parks. "Free Things" guide	Free social gathering spots
Social Indicators	CoM	"Feeling Lonely?" guide stats	Loneliness data by suburb/age
ABS Census 2021	ABS	Near Me: Demographic snapshot card	18–25 population by suburb

NHSD Services	Healthdirect	Near Me: Health + Mental Health	400K+ records, bulk-billing flag

3.3 Epics and User Stories

Epics are written in user story format (As a... I want... so that...). All acceptance criteria follow the Given/When/Then pattern.

Epic 1: Landing Page & Core Navigation [Must Have]

As a **young adult visiting Minuri for the first time**, I want to **see a clear homepage that explains what Minuri does and routes me to its features**, so that **I can quickly decide if it's relevant and start using it without confusion.**

User Story	US 1.1 — Homepage & Value Proposition
Description	As a young adult visiting Minuri for the first time, I want to see a clear homepage that explains what Minuri does and who it's for, so that I can decide if it's relevant to me.
Acceptance Criteria	AC1: Given I navigate to the Minuri homepage, when the page loads, then I see the Minuri branding, tagline ("Everything you need to know, but nobody taught you"), and a value proposition referencing Melbourne.
	AC2: Given I am on the homepage, when I scroll below the hero, then I see two persona cards (uni student and young professional) with descriptions linking to relevant guide categories.
	AC3: Given I am viewing the homepage on mobile (375px), when the page loads, then all content is visible with no horizontal scrolling and touch targets are at least 44px.
	AC4: Given I am on the homepage, when I look for actions, then I see two primary CTAs linking to First Time Guides and Near Me.

User Story	US 1.2 — Persistent Navigation
Description	As a user on any page, I want a persistent navigation bar, so that I can move between features without using the browser back button.
Acceptance Criteria	AC1: Given I am on any page, when I look at the top, then I see a navbar with links to: Home, Guides, Near Me, and Crisis Lines.
	AC2: Given I am on the Guides page, when I look at the navbar, then the Guides link is visually highlighted as active.
	AC3: Given I am on a screen below 768px, when I look at the navbar, then it collapses into a hamburger menu.
	AC4: Given I am on any page, when I look for crisis support, then I see a Crisis Lines link styled distinctly (red/orange accent) that is always visible.

Epic 2: First Time Guides [Must Have]

As a **young adult who has just moved to Melbourne**, I want to **browse and search practical guides on eating, transport, health, adulting, and social life**, so that **I can close the knowledge gap that nobody prepared me for when I left home.**

User Story	**US 2.1 — Guide Browsing by Category**
Description	As a young adult living alone for the first time, I want to browse guides by category, so that I can find relevant information without scrolling through everything.
Acceptance Criteria	AC1: Given I am on the Guides page, when the page loads, then I see a card grid with each card showing: title, category icon, reading time (2–5 min), and summary.
	AC2: Given I am on the Guides page, when I click “Eating & Cooking”, then only Eating & Cooking guides are displayed.
	AC3: Given I am viewing Guides on mobile, when the page loads, then cards stack vertically and are fully readable.

User Story	**US 2.2 — Guide Keyword Search**
Description	As a user, I want to search guides by keyword, so that I can find specific topics quickly.
Acceptance Criteria	AC1: Given I am on the Guides page, when I type “bulk billing” into the search bar, then only matching guides appear within 300ms.
	AC2: Given I search for a term with no matches, when results load, then I see “No results found.”
	AC3: Given I have a search query active, when I clear the search bar, then all guides reappear.

User Story	**US 2.3 — Guide Detail View with Data Widgets**
Description	As a user, I want to read a full guide with embedded data visualisations where relevant, so that I get practical advice and local information in one place.
Acceptance Criteria	AC1: Given I click the “Cheap Eats” guide, when the detail page loads, then I see the article content with an inline interactive map of cafés/restaurants from the CoM API.
	AC2: Given I am reading the “First GP” guide, when I reach the end, then I see a CTA “Find a GP near you →” linking to Near Me with the Health filter pre-selected.
	AC3: Given I am on a guide detail page, when I look at the bottom, then I see related guides from the same category.
	AC4: Given I click “Back to library”, when the Guides page loads, then my previous filter/search state is preserved.

User Story	**US 2.4 — Guide Bookmarking (No Login)**
Description	As a user, I want to bookmark guides, so that I can return to them later without creating an account.

Acceptance Criteria	AC1: Given I am on a guide card or detail page, when I click the bookmark icon, then the guide is saved to localStorage and the icon changes to a filled state.
	AC2: Given I have bookmarked 3 guides, when I select “My Bookmarks”, then only those 3 guides appear.
	AC3: Given I have bookmarked a guide, when I click the bookmark icon again, then it is removed and the icon reverts.
	AC4: Given I close and reopen the browser, when I visit the Guides page, then my bookmarks are still saved.

Epic 3: Near Me — Melbourne Resource Finder [Must Have]

As a **young adult who has just arrived in a Melbourne suburb**, I want to **see nearby health services, food options, parks, and amenities on a map**, so that **I know exactly what’s around me and can take action without searching multiple websites.**

User Story	**US 3.1 — Suburb Search & Service Results**
Description	As a young adult who just moved to a Melbourne suburb, I want to enter my suburb and see nearby services, so that I know where to eat, get help, and find amenities.
Acceptance Criteria	AC1: Given I am on Near Me, when I type “Clay”, then I see an autocomplete dropdown showing “Clayton” and other matching suburbs.
	AC2: Given I selected “Clayton”, when results load, then each result shows: name, address, type badge, and metadata (e.g. bulk-billing flag, seat count).
	AC3: Given I click “Use my location”, when the browser grants permission, then results load for my current suburb.

User Story	**US 3.2 — Suburb Demographic Snapshot**
Description	As a user, I want to see a demographic snapshot of my suburb, so that I understand how many young adults live here and what’s available.
Acceptance Criteria	AC1: Given I selected “Clayton”, when results load, then I see a card: “X young adults (18–25) live here” + service counts by category.
	AC2: Given the ETL script has run, when the suburb_demographics table is queried for Clayton, then it returns population_18_25 and total_population from ABS Census 2021.

User Story	**US 3.3 — Service Category Filtering**
Description	As a user, I want to filter services by type, so that I can find what I need without irrelevant results.
Acceptance Criteria	AC1: Given I am viewing results for Clayton, when I click “Health”, then only health services appear in the list and on the map.
	AC2: Given I have selected “Food”, when I also select “Parks”, then both categories display simultaneously.

	AC3: Given any filter is active, when I look at the page, then the Crisis Lines section is always visible.

User Story	US 3.4 — Interactive Map View
Description	As a user, I want to see services on an interactive map, so that I understand how far each one is from me.
Acceptance Criteria	AC1: Given I selected a suburb, when results load, then I see a Leaflet.js map with colour-coded pins for each service category.
	AC2: Given I am viewing the map, when I click a pin, then a popup shows: name, address, phone, and metadata.
	AC3: Given I selected a suburb, when the map renders, then it centres on that suburb with appropriate zoom.
	AC4: Given I am on mobile, when Near Me loads, then the map displays full-width below the results list.

Epic 4: Open Data Pipeline [Must Have]

As a **developer building Minuri**, I want to **ingest and proxy open data from ABS, CoM, and NHSD into a backend**, so that **the app delivers locally relevant, data-driven content without exposing external APIs to the frontend.**

User Story	US 4.1 — ABS Census Ingestion
Description	As a developer, I want ABS Census 2021 data ingested into PostgreSQL, so that Near Me can query suburb demographics.
Acceptance Criteria	AC1: Given the ABS CSV is downloaded, when I run ingest_census.py, then suburb_demographics is populated for all Greater Melbourne SA2 areas.
	AC2: Given the script has run before, when I run it again, then no duplicates are created.
	AC3: Given a request to /api/near-me/demographics?suburb=Clayton, when the API responds, then it returns population_18_25 and total_population.

User Story	US 4.2 — CoM API Proxy
Description	As a developer, I want a backend proxy for CoM API calls, so that the frontend never calls external APIs directly.
Acceptance Criteria	AC1: Given the frontend requests GET /api/near-me?suburb=Clayton&type=food, when the backend processes it, then it proxies to the CoM Cafés API and returns normalised results (name, type, address, lat, lng, metadata).
	AC2: Given the CoM API is unavailable, when cached results exist, then the backend serves cached data.
	AC3: Given a CoM API response is received, when the backend processes it, then the response is cached in PostgreSQL with a fetched_at timestamp.

User Story	US 4.3 — NHSD Health Integration
Description	As a developer, I want NHSD health service data integrated, so that users can find GPs and bulk-billing clinics.
Acceptance Criteria	AC1: Given the NHSD API key is configured, when a user requests type=health for postcode 3168, then results include name, address, phone, lat/lng, and bulk_billing flag.
	AC2: Given the NHSD key is unavailable, when a user requests type=health, then the backend falls back to CoM Landmarks (Health Services) + curated GP list.

Epic 5: Responsive Design & Accessibility [Should Have]

As a **mobile-first user**, I want to **have the entire app work well on my phone**, so that **I can use Minuri wherever I am without needing a desktop.**

User Story	US 5.1 — Mobile-First Responsiveness
Description	As a mobile user, I want Minuri to work well on my phone, so that I can use it on the go.
Acceptance Criteria	AC1: Given I view any page on 375px, when it loads, then all content is visible with no horizontal scrolling.
	AC2: Given I use a touch device, when I interact with any button or chip, then the touch target is at least 44px.
	AC3: Given I view Near Me on mobile, when it loads, then the map displays full-width below the list.

3.4 Technical Architecture

Frontend	React (TypeScript), Vite, Tailwind CSS, Axios, React Router. Leaflet.js for maps. localStorage for bookmarks.
Backend	FastAPI (Python), Pydantic, async. Proxy layer for CoM + NHSD APIs. REST endpoints for guides, suburbs, services, demographics.
Database	PostgreSQL (Supabase). Tables: articles, suburb_demographics, cached_places. SQLAlchemy + Alembic.
ETL	Python: ingest_census.py, seed_articles.py, cache_com_data.py. All idempotent.
External APIs	CoM: Opendatasoft v2.1 (no key). NHSD: Consumer API v5 (free key). ABS: CSV download.
Hosting	Frontend: Vercel. Backend: Railway / AWS EC2. DB: Supabase.
Auth	Not in Iteration 1. localStorage bookmarks. Auth planned for Iteration 2.

3.5 Features Deferred

Feature	Target	Rationale

Health Reminders	Iter 2	Requires persistent state + notifications. Without login, reminders reset each session.
Habit Tracker	Iter 2	Check-ins need cross-session persistence. localStorage loses data on clear.
Care Circle	Iter 2/3	Key differentiator. Requires auth, multi-user sharing, wellness scores. Better at higher weight.
First Time Navigator	Iter 3	Branching guided flows. Guides achieve similar knowledge-transfer with less risk.

Section B: LeanKit

Link to LeanKit: [Insert URL]

- **Backlog:** All epics with MoSCoW tags. Future: Reminders, Tracker, Care Circle, Navigator, Auth, Notifications.
- **To Do:** Epics 1–5, 14 user stories with Given/When/Then ACs.
- **Ready to Test:** Stories with ACs verified + test cases, awaiting mentor testing.
- **Done:** Implemented, tested, accepted.

Section C: Project Governance Portfolio

- Link to Design folder: [Insert link]
- Link to Data Governance folder: [Insert link]
- Link to Security Aspects folder: [Insert link]
- Link to Industry Mentor minutes: [Insert link]